MIL-D-50584 (MU)
30 September 1970
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

DELAY DETONATOR ASSEMBLY

(DWG. C-9207798)

1. SCOPE

1.1 This specification covers the Delay Detonator Assembly (Dwg. C-9207798).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-G-50427	-	Grenade, G.P.: M43A1, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1375

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MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. Army Munitions Command

F-9206731	-	Box, set-up, Packing, Ammo, Inner and Outer, for Delay Detonator Assembly
C-9206732	-	Box, Packing Ammunition, for Delay Detonator Assembly
C-9207798	-	Delay Detonator Assembly

PUBLICATIONS

ARMY

ET-9207798	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49	-	Transportation, Parts 171-179

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should specify, "49 CFR 171-179 (latest revision)".)

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3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with the applicable drawings and specifications.

3.2 Assemblies.-The delay detonator assemblies shall comply with all requirements specified on Drawing (Dwg.) C-9207798, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Grenade functioning.-The delay detonator assemblies shall function satisfactorily when subjected to this test in accordance with 4.4.3. (See 6.7)

3.4 Workmanship.-All delay detonator assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, projections, sharp edges, cracks, chips, dirt, grease, salt deposits, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material.

3.5 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.5).

3.6 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review. A statement of the findings on the moisture content for each explosive shall be furnished with each lot submitted.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1). The first article sample shall consist of one-hundred and thirty (130) delay detonator assemblies. All assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a

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change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies will be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings. In addition, the following examinations and tests shall be performed.

4.2.2.1 Static burning time (See Dwg. C-9207798).-Fifty (50) delay detonator assemblies shall be subjected to this test as specified in 4.4.2.

4.2.2.2 Grenade functioning (See 3.3).-Eighty (80) delay detonator assemblies shall be subjected to this test as specified in 4.4.3. (See 6.1.C)

4.2.3 Rejection.-If any assembly fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be con-

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sidered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Stems of one lot interfix number from one manufacturer.
b. Lead azide from one batch or blending of batches which results in homogeneity of all lead azide used in loading the entire detonator lot.
c. RDX from one batch or blending of batches which results in homogeneity of all RDX used in loading the entire detonator lot.
d. Delay composition from one batch or blending of batches which results in homogeneity of all delay compositions used in loading the entire detonator lot.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Delay detonator assembly (prior to loading lead azide and delay composition) (See Dwg. C-9207798).

Categories	Defects	Method of Inspection	Code Number (See 6.2)
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Weight of RDX	Balance	01001
102.	Diameter of shank, max	Gage	01002
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.4)	Visual	01003

4.3.2.2 Delay detonator assembly (See Dwg. C-9207798).

Categories	Defects	Method of Inspection	Code Number
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Diameter of flange, max	Gage	02001
102.	Charge flush to below flush, max	Gage	02002
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.4)	Visual	02003

4.3.2.3 Inner set-up box, prior to sealing (See Dwg. F-9206731).

Categories	Defects	Method of Inspection	Code Number
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Incorrect number of delay detonators	Visual	03001
102.	Packing cushion missing	Visual	03002
Minor:	None defined		

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4.3.2.4 Inner set-up box, sealed (See Dwg. F-9206731)

Categories	Defects	Method of Inspection	Code Numbe
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Box damaged to the extent that contents are exposed	Visual	04001
Minor	AQL 1.00 Percent		
201.	Contents loose	Visual	04002
202.	Marking misleading or unidentifiable	Visual	04003

4.3.2.5 Outer set-up box, prior to sealing (See Dwg. F-9206731).

Categories	Defects	Method of Inspection	Code Number
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Incorrect number of inner set-up boxes	Visual	05001
Minor	AQL 0.65 Percent		
201.	Filler material insufficient or missing	Visual	05002

4.3.2.6 Barrier bag, prior to sealing (See Dwg. F-9206731).

Categories	Defects	Method of Inspection	Code Number
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Corners not blunted	Visual	06001
102.	Sealing strip incomplete, badly wrinkled or missing	Visual	06002
Minor	AQL 0.65 Percent		
201.	Marking misleading or unidentifiable	Visual	06003

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4.3.2.7 Box, packing, wooden (prior to sealing) (See Dwg. C-9206732).

Categories	Defects	Method of Inspection	Code Number
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Incorrect number of barrier bags	Visual	07001
102.	Heat seals missing, unsealed or improperly positioned on barrier bags	Visual	07002
Minor	AQL 0.65 Percent		
201.	Marking misleading or unidentifiable	Visual	07003

4.3.2.8 Box, packing, wooden (sealed) (See Dwg. C-9206732).

Categories	Defects	Method of Inspection	Code Number
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Box damaged exposing contents	Visual-Manual	08001
102.	Strapping missing, broken or loose	Visual-Manual	08002
103.	Hardware missing broken or loose	Visual-Manual	08003
Minor	AQL 2.5 Percent		
201.	Strapping improperly assembled	Visual	08004
202.	Marking misleading or unidentifiable	Visual	08005
203.	Contents loose	Manual	08006
204.	Car seal missing, unsealed or improperly positioned	Visual	08007
205.	Hardware improperly engaged	Visual	08008

4.3.3 Testing.

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4.3.3.1 Moisture content of delay composition (See Dwg. C-9207798) Major defect (Code No. 09001).-The contractor shall provide adequate controls to insure compliance with the applicable requirement. The moisture content of each lot of delay composition shall be determined at the loading station at the time of loading the delay assemblies. This determination shall be made at least once for each eight (8) hours of production. A sample of approximately ten (10) grams shall be selected for this test. The sample shall be stored in an air tight container until this test is performed. If the moisture content fails to comply with the requirement specified on the applicable drawing and loading has not begun, the lot shall be rejected until corrective action has been verified by repeating the moisture content determination. If loading has begun, delay assemblies loaded with the non-conforming delay composition shall be rejected. This test shall be performed as specified in 4.4.1.

4.3.3.2 Static burning time (See Dwg. C-9207798) Major defect (Code No. 10001).-One-hundred and twenty-five (125) delay assemblies shall be selected from each lot at the loading station and be subjected to this test. If the average of the 125 sample assemblies fails to meet the requirement specified on the applicable drawing, or if any individual burning time is more than .220 seconds, or if more than one (1) individual time is less than .140 seconds, (the one (1) individual time may not be below .100 seconds), the lot shall be rejected. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.3.3 Grenade functioning (See 3.3) Major defect (Code No. 11001). Fifty (50) delay detonator assemblies shall be selected from each lot and subjected to this test. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4. If two (2) or more delay detonator assemblies fail to comply with specified requirement, the lot shall be rejected.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9207798 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacturer, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

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4.4 Test methods and procedures.

4.4.1 Moisture content of delay composition.-A tared, dried aluminum dish, 2 inches in diameter shall be covered with a sheet of teflon, 2½ inches square, and sufficiently flat to form a tight cover. An accurately weighed portion of approximately two grams shall be transferred to the moisture dish while both the cover and dish are on the balance pan. The dish and contents shall be transferred to an oven and dried at 95° C. plus or minus (±) 5° C. for 2 hours. The cover shall be placed in a sulphuric acid desiccator during the drying period. After drying, the dish shall be quickly transferred to the desiccator and allowed to cool to room temperature. The teflon cover shall be placed on the dish and assembly weighed. The loss in weight shall be calculated to percentage of moisture in the delay composition as follows:

$$\text{Percentage of moisture} = \frac{(A - B) \times 100}{A}$$

where

A = The weight of the specimen before drying.
B = The weight of the specimen after drying.

4.4.2 Static burning time.-This test shall be performed using the procedure and equipment specified on the equipment list. Observation shall be made to determine compliance with the static burning time requirements specified on the applicable drawing.

4.4.3 Grenade functioning test of delay detonator assembly (See 3.3 and MIL-G-50427) Major defect (Code No. 11001).-The sample delay detonator assemblies shall be assembled in grenades and subjected to this test in accordance with MIL-G-50427, paragraph 4.4.2. In addition, partial duds shall be disassembled in order to determine whether the delay detonator assembly functioned properly. All information recorded shall be

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forwarded as specified in 6.5. Malfunctions not attributed to the delay detonator assembly shall be disregarded. Provision shall be made for replacement of such grenades.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (Level C).-The delay detonator assemblies shall be preserved and packaged in accordance with Dwg. F-9206731.

5.2 Packing (Level C).-The delay detonator assemblies, preserved and packaged as specified in 5.1, shall be packed in accordance with Dwg. C-9206732.

5.3 Marking.-Marking shall be in accordance with Dwgs. F-9206731 and C-9206732 and Code of Federal Regulations, Title 49, Parts 171-179.

5.4 Shipping.-When delay detonator assemblies from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples. The procuring agency shall specify the facility at which the grenade functioning test shall be conducted.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (See 6.3.5).

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6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (See 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category A, Form 3. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Check test for possible deterioration.-The following test is designed to prevent delay detonator assemblies that have deteriorated from being assembled into ammunition items. The test shall be performed on delay detonator assemblies from each lot which have been stored for more than two years or under adverse conditions for any period of time. If the specification

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for the ammunition item into which the delay detonator assemblies are assembled does not require that the check test be performed, the contracting officer should insert in the contract a provision for the performance of the check test. The contracting officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. One-hundred and twenty-five (125) delay detonator assemblies shall be selected from each lot and tested as follows:

6.4.1 Static burning time.-The delay detonator assemblies shall be tested as specified in 4.4.2. If the average burning time of the 125 delay detonator assemblies fails to meet the requirement specified on Dwg. C-9207798, or if two (2) or more individual burning times are more than 0.220 seconds, or if three (3) or more burning times are less than 0.140 seconds, the lot shall be rejected and not loaded into the end item. No individual burning time shall be less than 0.100 seconds.

6.5 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

6.6 Supersession data.-This specification includes the requirements of, and supersedes, Purchase Specification AEI-304, dated 24 November 1965.

6.7 Satisfactory functioning.-Satisfactory functioning of the delay detonator assembly shall be defined as complete burning of the delay composition, as indicated by burning within the delay time specified on the applicable drawing, and subsequent high order propogation of the lead azide and RDX.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1375-A117

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